Cart Abandonment Pattern Analysis

Python + SQL + Power BI | 55M events | 1.66M cart sessions | Feb 2020

VICKY JEERU
DATA ANALYST | POWER BI DEVELOPER
Hyderabad, India
jeeruvicky51@gmail.com
+91 91541 39298
LinkedIn **Portfolio** **GitHub**

55.3M
events processed in chunks to isolate 1.66M cart sessions

57.8%
cart abandonment: 960K of 1.66M cart sessions ended without a purchase

89.4%
of cart sessions hold a single product, and 60.2% of them are abandoned

-19.5 pts
abandonment for repeat users (6+ cart sessions) vs one-time users

Executive Summary

More than half of the customers who put something in their cart never buy it. Using **Python, SQL (DuckDB) and Power BI**, I processed the **7.15 GB February 2020 clickstream file (55.3 million events)** in chunks, built a clean cart-session dataset, answered ten business questions in SQL, and published a four-page interactive dashboard.

> **Bottom line:** cart abandonment is **57.8%**, and it is driven less by *what* is in the cart than by *who* is buying and *how engaged* they are. One-time visitors abandon 65.4% of carts versus 45.9% for repeat users, single-product carts (89% of all carts) abandon 60.2% versus about 30% for carts with 4+ products, and sessions that stay within a single minute abandon 63.9%. Each 1-point cut in abandonment is worth about **16,600 extra purchase sessions** a month.

Key Findings at a Glance

- **User familiarity:** abandonment falls from 65.4% (one cart session) to 45.9% (6+ cart sessions).
- **Cart depth and engagement:** carts with 2+ products abandon 30-40%; sessions with 3-10 minutes of cart activity abandon about 30%, versus 63.9% for single-moment sessions (78% of sessions).
- **Price:** items under $50 abandon 65.6% of the time versus about 50% for $100-350 items.
- **Timing:** abandonment is about 56% from 05:00-12:00 UTC and 59-62% from 14:00-18:00 UTC.
- **Categories and brands:** extreme rates (smartphones 93.8%, Silverlit 97.8%) come from small volumes; among high-volume items, bicycles, shoes, clocks, Sony and Bosch sit 3-6 points above average.

Business Problem

Why do customers abandon their carts, and which customer, product and timing factors are most strongly associated with it? The goal is to tell an e-commerce product team where to investigate first, using patterns that are real and large enough to act on.

Tools & Techniques

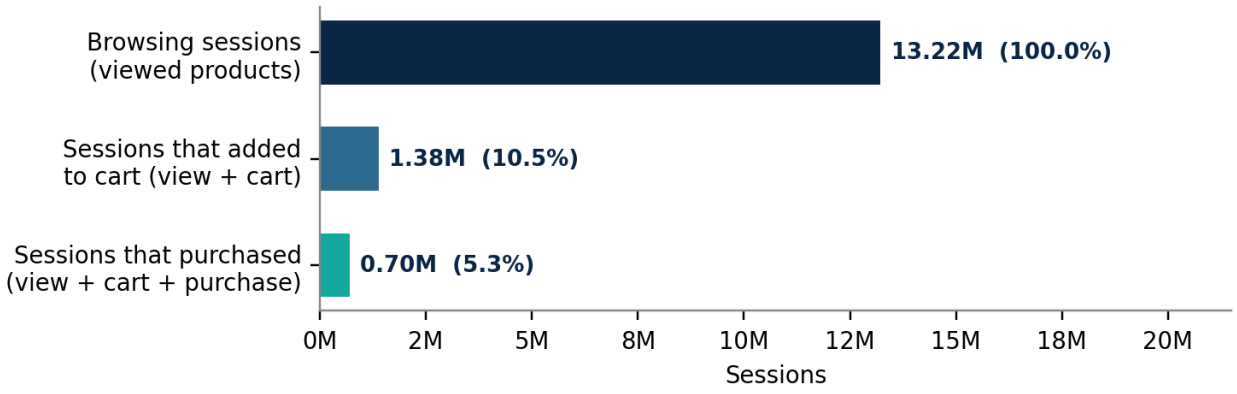
Area	What I used
Python	Pandas chunked processing (500K rows per chunk), set operations on session IDs, de-duplication, price banding, repeat-session analysis
SQL (DuckDB)	10 business questions using CTEs, `COUNT(DISTINCT)`, CASE banding, `DATE_DIFF`, `EXTRACT` and volume filters
Power BI	4-page interactive dashboard with KPI cards, slicers, funnel, and driver charts

1. Data, Method & the Purchase Funnel

The source is the Kaggle *eCommerce Behavior Data from Multi-Category Store*, February 2020. The raw file is an event log (view, cart, purchase), so it had to be turned into a cart-session table before any analysis.

Step	Result	Why it matters
1. Scan full file in chunks	55,318,565 events: 51.2M views, 2.89M carts, 1.20M purchases; 4.23M users; 13.5M sessions	7.15 GB cannot be loaded at once; a 1,000-row sample was used only to learn the schema
2. Keep cart events	2,885,608 cart events, each flagged `purchased` if its session also contains a purchase	Defines abandonment at session level: cart without a purchase
3. Remove exact duplicates	229,080 duplicates (7.9%) removed, leaving 2,656,528 cart rows	Repeated events would overstate cart activity
4. Handle missing values	Category missing on 203,714 rows (7.67%), brand on 10%: kept and labelled "Unknown"	A missing label is still a real cart event; deleting it would bias results
5. Add time features	Date, hour (UTC), weekday from the event timestamp	Enables timing analysis

The purchase funnel



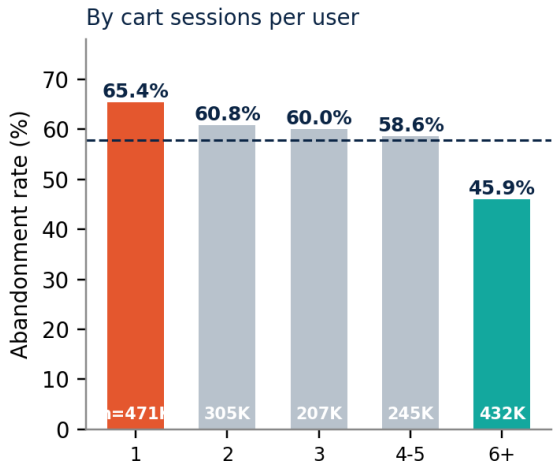
Stage	Sessions	Share of previous stage
Browsing (viewed products)	13,217,465	100%
Added to cart (view + cart)	1,383,989	10.5%
Purchased (view + cart + purchase)	698,497	50.5%

Two different drop-offs: 89.5% of browsing sessions never reach the cart, and 57.8% of cart sessions never buy. Cart abandonment is the *more actionable* loss because these customers already showed purchase intent, but in percentage terms the larger drop is earlier, between viewing and adding to cart.

Headline numbers (all cart sessions: 1,660,110): **700,091 purchased (42.17%)** and **960,019 abandoned (57.83%)**. SQL counts one fewer session (1,660,109) because a single null session ID is ignored by `COUNT(DISTINCT)`; the rates are unchanged.

2. Customer & Session Behavior

Finding 1: Repeat users abandon far less

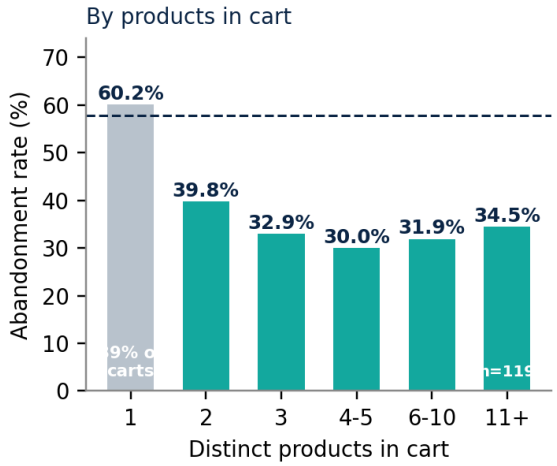


Users are grouped by how many **cart sessions** they had in February. One-time users abandon **65.4%** of carts; users with 6+ cart sessions abandon just **45.9%**, a **19.5-point** gap.

- One-time users are **28%** of cart sessions (470,701 sessions); 6+ session users are 26%
- Abandonment drops steadily: 65.4%, 60.8%, 60.0%, 58.6%, then 45.9%

Repeat buyers may simply return more often, so this shows association, not cause.

Finding 2: Bigger carts convert better

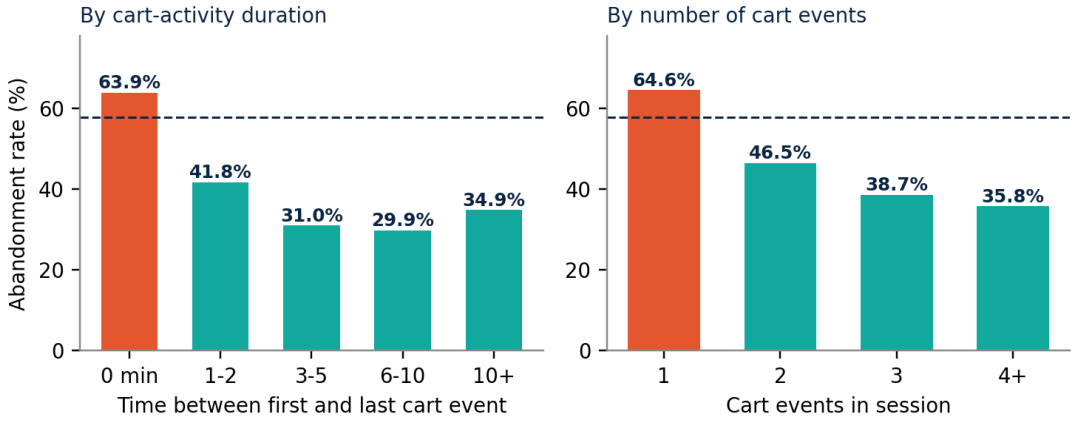


89.4% of cart sessions (1,484,011) contain just one product, and they abandon **60.2%** of the time. Adding a second product drops abandonment to **39.8%**, and 4-5 products to **30.0%**.

- Purchasers average 1.21 distinct products vs 1.09 for abandoners
- The 11+ product group has only 119 sessions, so ignore its rate

Customers with deeper carts show stronger purchase intent, which makes the single-product cart the largest target segment.

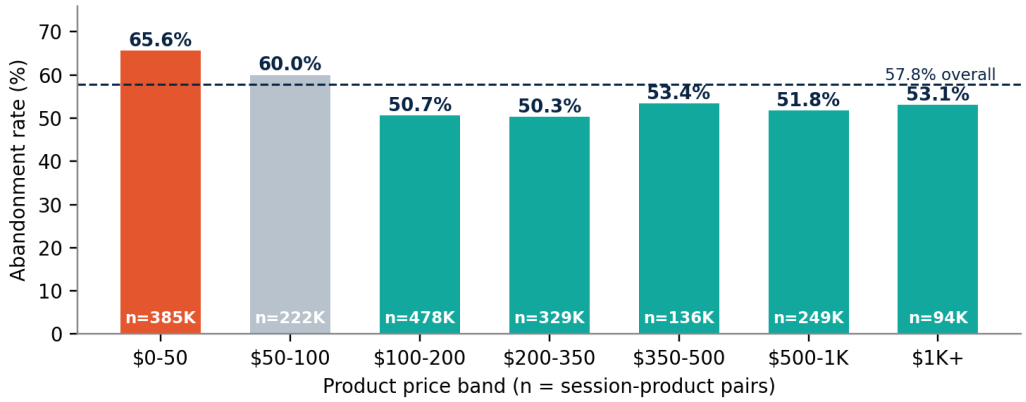
Finding 3: Instant sessions rarely convert



- **77.9% of cart sessions (1,292,378) stay within a single minute** between first and last cart event, and 63.9% of them are abandoned. Sessions with 3-10 minutes of cart activity abandon only 30-31%.
- Purchasers average **1.85 cart events** vs 1.42 for abandoners; sessions with 4+ cart events abandon 35.8% vs 64.6% for a single event.

3. Price, Category & Brand

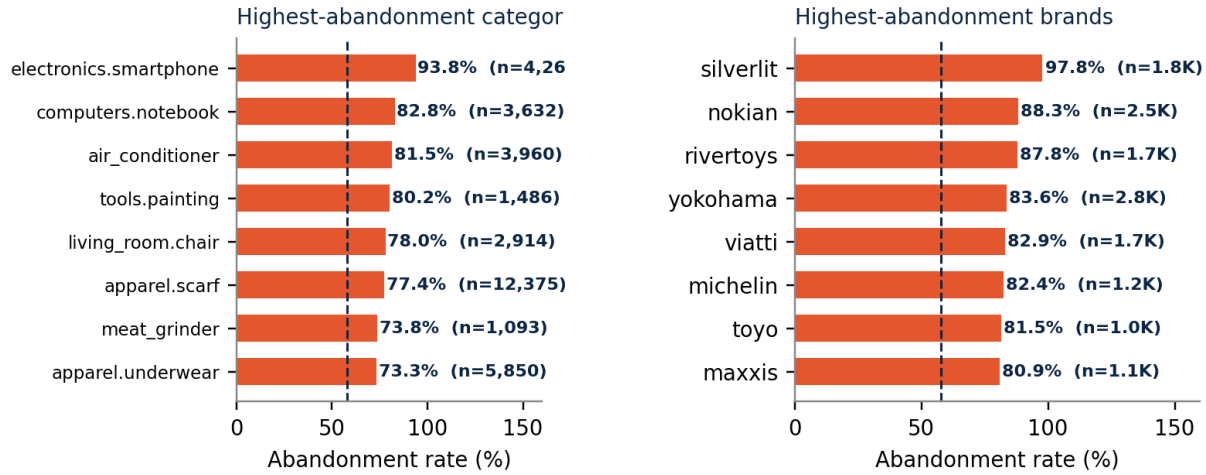
Finding 4: Low-priced items are abandoned most



Items under $50 are abandoned **65.6%** of the time and $50-100 items **60.0%**, versus roughly **50-53%** for everything above $100. The overall median cart price is $172.44, so the high-abandonment zone is the cheap end of the catalog.

Insight: a fixed shipping fee or minimum-order rule weighs more on a small basket, so test shipping thresholds and bundles on sub-$100 carts first.

Finding 5: Category and brand: read volume before rate

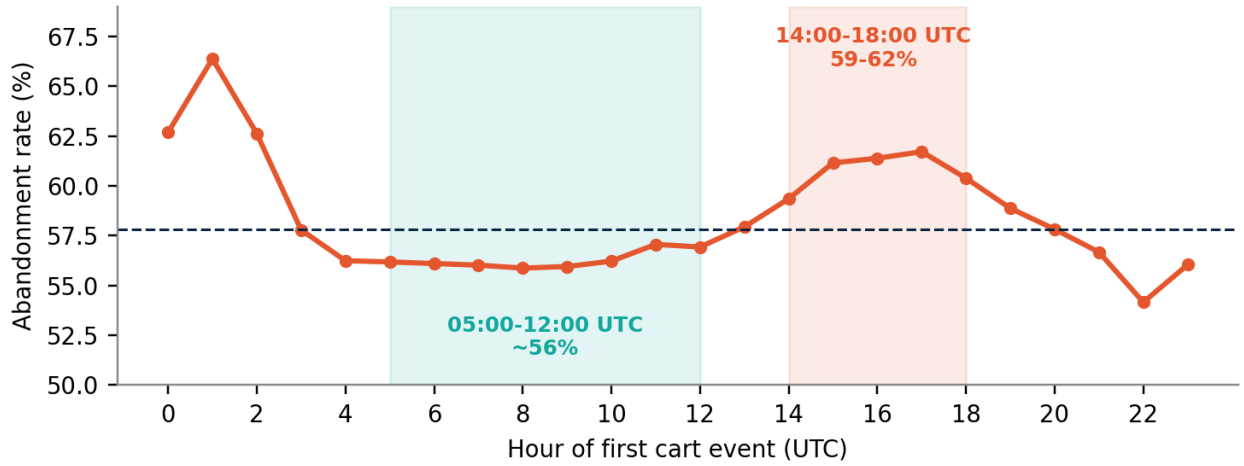


Largest categories (sessions)	Abandonment	Largest brands (sessions)	Abandonment
construction.tools.light (657K)	47.6%	Apple (354K)	54.9%
Unknown / no category (139K)	68.9%	Samsung (305K)	45.3%
sport.bicycle (118K)	61.3%	Xiaomi (154K)	55.2%
electronics.clocks (64K)	61.1%	Sony (19K)	61.0%
apparel.shoes (56K)	63.5%	Bosch (14K)	62.4%

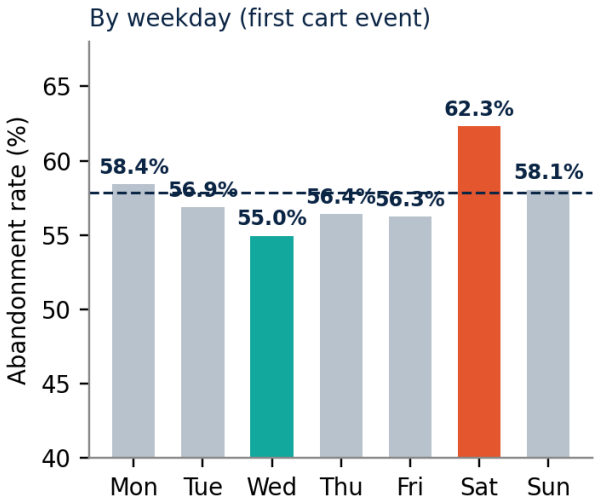
- The highest rates (smartphones 93.8%, Silverlit 97.8%) come from only **1,000-5,000 sessions**; many top brands are tire brands, so brand partly reflects product type.
- Among **high-volume** items, Unknown category (68.9%), shoes (63.5%), bicycles (61.3%), Sony (61.0%) and Bosch (62.4%) run above the 57.8% average; Samsung (45.3%) and Oppo (40.8%) run far below.
- **Data caution:** `construction.tools.light` holds 40% of carts (47.6% abandonment), yet Apple and Samsung items appear under it in the sample, so the label looks mis-coded.

4. Timing & the Dashboard

Finding 6: Time of day matters a little



Abandonment is lowest (about **56%**) between 05:00 and 12:00 UTC and highest (**59-62%**) between 14:00 and 18:00 UTC, a 5-6 point swing. Cart volume peaks in the morning window, so the afternoon rise affects fewer, but less committed, shoppers.



Weekday looks different, but treat it carefully. Saturday shows the highest abandonment (62.3%) and Wednesday the lowest (55.0%).

- 1 February 2020 was a Saturday, and its daily abandonment on the dashboard is **74.5%** on about 104K sessions, which by itself lifts Saturdays; the other Saturdays average roughly 56%, in line with other days
- The dashboard also shows an outage-like dip to about 4K sessions with an 86% abandonment spike around 26-27 February

Both look like data-boundary or tracking effects, so weekday and daily-trend results need checking before any decision.

Power BI dashboard (4 pages)

Page	Content
Executive Overview	KPI cards (cart sessions, abandoned, abandonment rate, purchase rate, purchased), Category / Date / Brand slicers, daily cart sessions and daily abandonment trends
Funnel Analysis	Cart-to-purchase funnel (1.66M to 0.70M, 42.2%), abandonment by hour of day, key takeaway and "what should the product team fix first"
Abandonment Drivers	Highest-abandonment categories and brands, abandonment by price band and by product count
Customer & Session Behavior	Purchasers vs abandoners by product count, product-count distribution, purchasers vs abandoners donut, key insights

5. Recommendations, Limitations & Next Steps

Recommendations (in priority order)

#	Action	Evidence
1	**Test cross-sell on single-product carts:** related-item recommendations and free-shipping-threshold nudges	89.4% of carts; 60.2% abandon vs 39.8% for 2 products
2	**Win back one-time visitors:** guest-friendly checkout, trust signals, cart-reminder emails or push	65.4% abandonment vs 45.9% for repeat users; 28% of cart sessions
3	**Re-engage instant sessions:** capture email or save cart before the visitor leaves	77.9% of sessions stay within one minute; 63.9% abandon
4	**Review sub-$100 baskets:** shipping fees, minimum order, bundles	65.6% ($0-50) and 60.0% ($50-100) vs about 50% above $100
5	**Fix catalog data, then review high-volume weak spots:** 7.7% of carts have no category; check shoes, bicycles, Sony, Bosch and smartphones	Unknown 68.9%, shoes 63.5%, bicycles 61.3%, smartphones 93.8% (n = 4,263)
6	**Schedule recovery campaigns for 14:00-18:00 UTC** when abandonment peaks	59-62% vs about 56% in the morning

Sizing: every 1-point reduction in abandonment equals about **16,600 additional purchase sessions** per month (1% of 1.66M cart sessions).

Limitations

- **Association, not causation:** repeat buyers may return more often, and deeper carts may reflect existing intent.
- **No checkout-step, device or traffic-source fields** exist in this dataset, so the exact friction point, device and channel effects cannot be measured here.
- **One month of data (February 2020)** with start-of-month and late-February anomalies; price and brand results use session-product counts and session counts respectively.

Next steps

- Instrument checkout steps, device type and traffic source, then rerun the analysis by channel.
- Add an A/B test plan for recommendations 1-3 with abandonment rate as the success metric.
- Compare several months and exclude data-boundary days to confirm the weekday and daily patterns.

Vicky Jeeru | Data Analyst & Power BI Developer | Hyderabad, India
jeeruvicky51@gmail.com | +91 91541 39298 | GitHub | Portfolio | LinkedIn
Code and dashboard: https://github.com/jeeruvicky51-bit/Cart-Abandonment-Pattern-Analysis